JavaScript Fundamentals

1. What JavaScript is JavaScript is the programming language of the browser

HTML describes the structure of a page and CSS describes its appearance, while JavaScript describes its behaviour: what happens when a user clicks, types, scrolls or submits a form. Modern JavaScript also runs outside the browser on servers through Node.js.

2. Variables There are three ways to declare a variable

var is the oldest form and is function-scoped, which often surprises beginners. let declares a block-scoped variable that can be reassigned. const also declares a block-scoped variable, but the binding cannot be reassigned after it is created. An object declared with const can still have its properties changed; only the binding is fixed.

3. Data types and coercion JavaScript has seven primitive types: string, number, boolean, null, undefined, symbol and bigint

Everything else is an object, including arrays and functions. JavaScript converts types automatically in many expressions, which is called coercion. For example, the expression '5' + 3 produces the string '53' because the plus operator prefers string concatenation when one operand is a string. The expression '5' - 3 produces the number 2 because the minus operator only works on numbers. Use === instead of == to compare without coercion.

4. Functions A function is a reusable block of code

A function declaration is hoisted, meaning it can be called before it appears in the file. A function expression assigned to a const is not hoisted in the same way. Arrow functions, written as (a, b) => a + b, are shorter and do not bind their own this value, which makes them convenient as callbacks.

5. The DOM The Document Object Model is a tree of objects that the browser builds from the HTML document

Each element becomes a node that JavaScript can read and change. document.querySelector finds the first element matching a CSS selector, and textContent or classList can then be used to update it. Changing the DOM changes what the user sees immediately, without reloading the page.

6. Events An event is something that happens in the page, such as a click, a keypress or a form submission

addEventListener registers a handler function that runs when a named event occurs on an element. The handler receives an event object with details such as which element was the target. event.preventDefault() stops the browser's default behaviour, which is commonly used to stop a form from reloading the page.

7. Arrays and iteration Arrays store ordered lists

The most useful methods are map, which returns a new array with each item transformed, filter, which returns a new array with only the items that pass a test, and reduce, which collapses an array into a single value. These methods do not modify the original array, which makes code easier to reason about.

8. Asynchronous JavaScript JavaScript runs on a single thread, so slow work such as a network request must be asynchronous

A Promise represents a value that will be available later. fetch returns a Promise for an HTTP response. The async and await keywords let asynchronous code be written in a sequential style: await pauses inside an async function until the Promise settles, without blocking the rest of the page. Errors from awaited calls are caught with try and catch.

9. JSON JSON is a text format for structured data and is how browsers and servers usually exchange information

JSON.stringify converts a JavaScript value into a JSON string, and JSON.parse turns a JSON string back into a JavaScript value. A fetch response has a .json() method that reads the body and parses it.

10. Common beginner mistakes Comparing with == instead of ===, forgetting that array methods like map return a new array instead of changing the original, assuming code after an asynchronous call has already finished, and querying the DOM before the elements exist in the page.